



Figure 4: Cryptocurrency trading platforms (Binance) are vulnerable to hacks (Source: CoinTelegraph)

Flooding the network with low-cost, high-complexity transactions is one way of messing with the Ethereum blockchain. Other than introducing additional transactions, the blockchain is also bloated by adding in multiple empty accounts. An empty account can be created on Ethereum by transferring zero *Ether* to an address. Since the account has no smart contract code, no stored data, and no *Ether* to send to other accounts, it contains nothing of value. Yet, since a transaction was made to the account, the result of this is stored on the blockchain, increasing its size. This delays other transactions. Since a transaction is fairly cheap in *gas*, the hacker(s) took advantage of this to attack the blockchain and significantly bloat it. The Ethereum developers responded by taking measures aimed at debloating the blockchain, and introduced additional protections against DDoS threats.

Decentralised applications

Ethereum made its way through the onslaught of attacks and emerged a powerful platform with significant potential for building decentralised applications or dApps.

According to BlockGeeks, the following key aspects of dApps need to be focused on, particularly in the light of their practical application in other domains:

1. **Open source:** The decentralised application should be made open to all scrutiny and governed by autonomy. Decisions on changes should be made by the consensus of a majority of the stakeholders.

2. **Decentralised:** Underscoring what the name dApp stands for, the records relating to transactions between two or more parties should be stored on a public ledger using the blockchain.
3. **Incentives:** Entities or individuals involved in validating the blockchain should be appropriately rewarded for their work. In most cases, this is done by offering them cryptographic tokens for the currency or the application in question.
4. **Protocol:** The network should agree on a cryptographic algorithm that shows the proof of value. For instance, the Bitcoin network follows the Proof of Work model, while Ethereum is slated to move towards Proof of Stake entirely. Some areas in which dApps

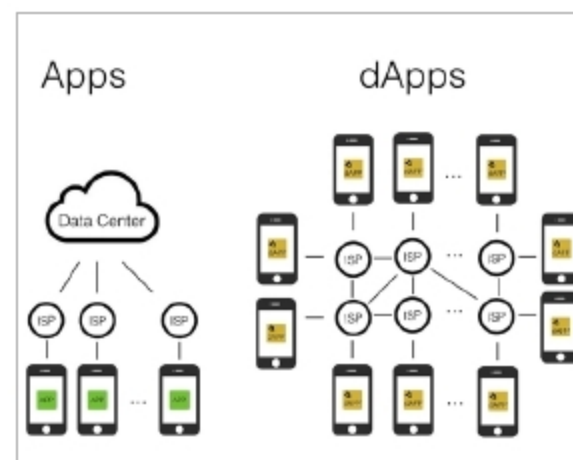


Figure 5: Regular apps versus decentralised applications (Source: Shaan Ray)

can drive a paradigm shift include healthcare, security, digital finance, politics, data storage and gambling. In fact, gambling is one of the top areas in which dApps are operating. Given that it is over a US\$ 200 billion market, making gambling transparent might lead to some 'game-changing' new developments for players, particularly in the online arena that has not traditionally seen this level of openness in transactions.

The amount of fake news being generated is on the rise. Using the blockchain to verify the source of this news may well help identify bots and sources of misinformation that can otherwise influence large swathes of the populace before coming under scrutiny. Votes can also be registered via the blockchain, which could dramatically cut down on fake votes and even open up opportunities to transfer the voting process entirely online in a decade or two. This could tie in with the idea of each individual possessing a national digital identity. Being in charge of this publicly visible persona could reduce the chances of identity theft, since there is then a quick and concrete way to identify real ownership.

There have been a number of data leaks in the past few years across the technology sector, and in most cases this was due to insecure centralised systems of data storage. In contrast, introducing internal blockchains to store customer data might allow companies to transfer the processing to decentralised data stores, thereby reducing the damage from any single point of failure that attackers could exploit.

A number of different cryptocurrencies have actually used the Ethereum platform to launch their initial coin offerings. Clearly, there is huge potential for the technology and, as Ethereum states, 'to create a decentralised Internet'. It remains to be seen where we go with the marvels this platform has to offer and whether it blossoms into a game-changer for the centralised infrastructure we rely on today, or gets forgotten as a has-been by-product of the dynamic world of technology. **END** 🐧

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